

Upper Bounds on the 2-rainbow Domination Number of Grids*

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Abstract

The k -rainbow domination is a variant of the classical domination problem in graphs and is defined as follows. Given an undirected graph $G = (V, E)$, we have a set of k colors and we assign an arbitrary subset of these colors to each vertex of G . If a vertex which is assigned an empty set, then the union of color set of its neighbors must be k colors. This assignment is called the k -rainbow dominating function of G . The minimum sum of numbers of assigned colors over all vertices of G , is called the k -rainbow domination number of G . In this paper, we present upper bounds on the 2-rainbow domination number of Grids.

Keywords: Domination, 2-rainbow domination, Grids

1. Introduction

The k -rainbow domination is a variant of classical domination [7, 8]. Let $G = (V(G), E(G))$ be a finite, simple and undirected graph, where $V(G)$ and $E(G)$ are the vertex and edge sets of G , respectively. For a vertex $v \in V(G)$, the open neighborhood $N(v) = \{u \in V(G) \mid (u, v) \in E(G)\}$. Let $C = \{1, 2, \dots, k\}$ be a set of k colors, and f be a function that assign to each vertex a set of colors chosen from C , that is, $f: V(G) \rightarrow \mathcal{P}(\{1, \dots, k\})$. If for each vertex $v \in V(G)$ such that $f(v) = \emptyset$ we have $\bigcup_{u \in N(v)} f(u) = C$ then f is called a k -rainbow dominating function (kRDF) of G . The weight, $\omega(f)$, of a function f is defined as $\omega(f) = \sum_{v \in V(G)} |f(v)|$. The k -rainbow domination number $\gamma_{rk}(G)$ of G is the minimum weight of a k -rainbow dominating function. For example, a 2-rainbow domination of path P_5 is shown in Figure 1(a). We assign a color set $\{1\}$ to V_1, V_5 and assign a color set $\{2\}$ to V_3 . Since $f(V_2) = \emptyset$ and $f(V_4) = \emptyset$, $\bigcup_{u \in N(V_2)} f(u)$ and $\bigcup_{u \in N(V_4)} f(u)$ must be $\{1, 2\}$. In fact, $\gamma_{r2}(P_5) = 3$ by Observation 1. A 2-rainbow domination of path P_4 is shown in Figure 1(b). We assign a color set $\{1, 2\}$ to V_2 and assign a color set $\{1\}$ to V_4 . Similarly, the union of color set of neighbors of V_1 and V_3 must be $\{1, 2\}$. By Observation 1, $\gamma_{r2}(P_4) = 3$.

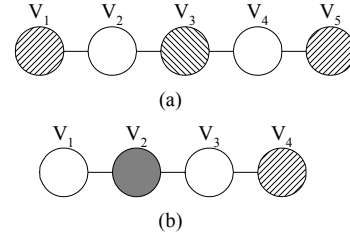


Figure 1: 2-rainbow dominations of P_5 and P_4 . The vertex with a color set $\{1, 2\}$ is represented by a gray vertex, and the vertex with color sets $\{1\}$ and $\{2\}$ are filled with slashes and backslashes, respectively.

In [2, 3], Brešar, Henning and Rall introduced the concept of k -rainbow domination of a graph G . In [4], Brešar and Šumenjak showed that the decision version of 2-rainbow domination of graphs is NP-complete even when restricted to chordal graphs (or bipartite graphs). They also gave the exact values of the 2-rainbow domination numbers for paths, cycles and suns. Thus, further investigations tended to study bounds on k -rainbow domination number for certain families of graphs, such as trees and generalized Petersen graphs. This problem is widely studied in [1-6, 9-13].

For $m \geq n \geq 1$, the *Grid*, $G_{n,m}$, is a graph with vertex set $V(G_{n,m}) = \{(x, y) \mid 1 \leq x \leq n \text{ and } 1 \leq y \leq m\}$ and two vertices (x_1, y_1) and (x_2, y_2) are adjacent if and only if $|x_1 - x_2| + |y_1 - y_2| = 1$. For example, Figure 2 depicts $G_{3,4}$. In this paper, we present upper bounds of $\gamma_{r2}(G_{n,m})$.

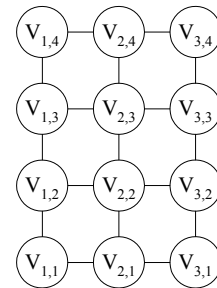


Figure 2: $G_{3,4}$.

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2. Main Result

All graphs considered here are undirected and simple (i.e., finite, loopless, and without multiple edges). Let $G = (V(G), E(G))$ be a graph with vertex set $V(G)$ and edge set $E(G)$.

Observation 1 (Brešar et al. [4]). $\gamma_{r2}(P_n) = \lfloor n/2 \rfloor + 1$.

Since $G_{1,m}$ is isomorphic to P_n , we immediately obtain the following corollary.

Corollary 2. $\gamma_{r2}(G_{1,m}) = \lfloor m/2 \rfloor + 1$ for $m \geq 1$.

Lemma 3 $\gamma_{r2}(G_{n,m}) \leq \lfloor n \times m/2 \rfloor$ for $2 \leq n \leq 4$ and $m \geq n$.

Proof. To achieve this upper bound, we give a color assignment method. Let $v_{i,j}$ be assigned a color set $\{1\}$ for $1 \leq \text{odd integer } i \leq n$ and $2 \leq \text{even integer } j \leq m$. Let $v_{i,j}$ be assigned a color set $\{2\}$ for $2 \leq \text{even integer } i \leq n$ and $1 \leq \text{odd integer } j \leq m$. It's easy to verify that $\bigcup_{u \in N(v)} f(u) = \{1, 2\}$ which $f(v) = \emptyset$. $\square$

For example, Figure 3(a) and 3(b) show 2-rainbow dominations of $G_{3,5}$ and $G_{4,4}$, respectively.

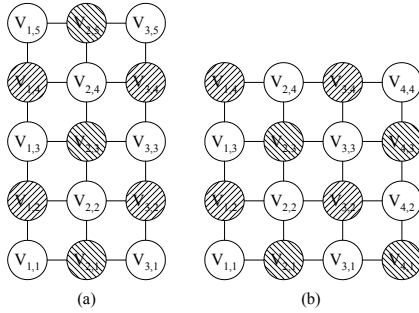


Figure 3: 2-rainbow dominations of $G_{3,5}$ and $G_{4,4}$. The vertex with color sets $\{1\}$ and $\{2\}$ are filled with slashes and backslashes, respectively.

Lemma 4 $\gamma_{r2}(G_{4i+1,4j+1}) \leq 6ij + 2i + 2j$ for both i and j are integer and $j \geq i \geq 1$.

Proof. To achieve this upper bound, we give a color assignment method. We consider 2-rainbow dominations of $G_{5,5}$ first. Let Pattern A be a color assignment of $G_{5,5}$ which is consist of $V_{3,3}$ with a color set $\{1, 2\}$, vertices $V_{1,4}, V_{2,1}, V_{4,5}, V_{5,2}$ with a color set $\{1\}$ and vertices $V_{1,2}, V_{2,5}, V_{4,1}, V_{5,4}$ with a color set $\{2\}$. Let Pattern B be a color assignment of $G_{5,5}$ which is consist of $V_{3,3}$ with a color set $\{1, 2\}$, vertices $V_{1,2}, V_{2,5}, V_{4,1}, V_{5,4}$ with a color set $\{1\}$ and vertices $V_{1,4}, V_{2,1}, V_{4,5}, V_{5,2}$ with a color set $\{2\}$. Figure 4(a) and 4(b) shows Pattern A and B, respectively. It's easy to verify that both Pattern A and B are 2-rainbow dominations of $G_{5,5}$.

Then, we consider 2-rainbow dominations of $G_{5,4j+1}$ for $j \geq 2$. We give a color assignment which is

consist of alternate Pattern A and B and the last row of k -th Pattern is overlapping to the first row of $(k+1)$ -th Pattern for $1 \leq k < j$.

The color assignment of $G_{4i+1,4j+1}$ for $j \geq i \geq 2$ is similar to previous period. For $i \geq 2$, we give a color assignment which is consist of alternate Pattern A and B and the last column of k -th Pattern is overlapping to the first column of $(k+1)$ -th Pattern for $1 \leq k < i$.

Sum of numbers of assigned colors over all vertices of $G_{4i+1,4j+1}$ is $10ij - 2(i-1) - 2(j-1) - 4(i-1)(j-1)$. After computing, we have $6ij + 2i + 2j$. $\square$

For example, Figure 5 shows a 2-rainbow domination of $G_{9,9}$ which is consist of alternate Pattern A and B.

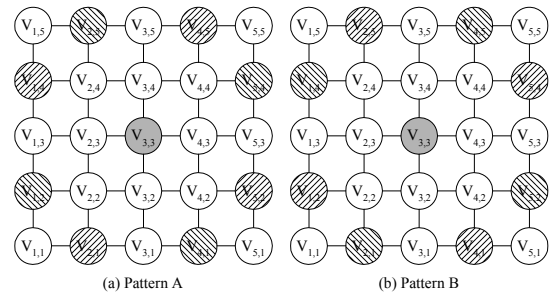


Figure 4: 2-rainbow dominations of $G_{5,5}$. The vertex with a color set $\{1, 2\}$ is represented by a gray vertex, and the vertex with color sets $\{1\}$ and $\{2\}$ are filled with slashes and backslashes, respectively.

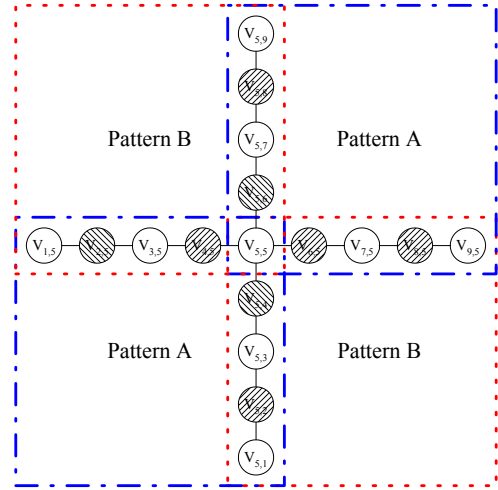


Figure 5: A 2-rainbow dominations of $G_{9,9}$. The vertex with a color set $\{1, 2\}$ is represented by a gray vertex, and the vertex with color sets $\{1\}$ and $\{2\}$ are filled with slashes and backslashes, respectively.

Lemma 5 $\gamma_{r2}(G_{n,m}) \leq 6\lfloor (n-1)/4 \rfloor \lfloor (m-1)/4 \rfloor + 2\lfloor (n-1)/4 \rfloor + 2\lfloor (m-1)/4 \rfloor + (m+n)/2$ for $n \equiv 2 \pmod{4}$ and $m \equiv 2 \pmod{4}$, $m \geq n \geq 5$.

Proof. We partition $G_{n,m}$ to a $G_{4i+1,4j+1}$ and the remain part which $i = \lfloor (n-1)/4 \rfloor$ and $j = \lfloor (m-1)/4 \rfloor$. In fact, the

remaining part is a Path P_{n+m-1} . Combining the results of Observation 1 and Lemma 4, we immediately obtain the upper bound $6\lfloor(n-1)/4\rfloor \lfloor(m-1)/4\rfloor + 2\lfloor(n-1)/4\rfloor + 2\lfloor(m-1)/4\rfloor + (m+n)/2$. $\square$

For example, Figure 6 shows a 2-rainbow domination of $G_{6,6}$ which can be partitioned to a $G_{5,5}$ and a Path P_{11} .

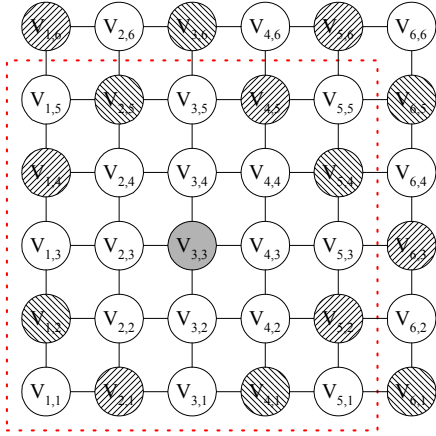


Figure 6: A 2-rainbow domination of $G_{6,6}$. The vertex with a color set $\{1, 2\}$ is represented by a gray vertex, and the vertex with color sets $\{1\}$ and $\{2\}$ are filled with slashes and backslashes, respectively.

Lemma 6 $\gamma_{r2}(G_{n,m}) \leq 6\lfloor(n-1)/4\rfloor \lfloor(m-1)/4\rfloor + 2\lfloor(n-1)/4\rfloor + 2\lfloor(m-1)/4\rfloor + 3(m+n)/2 - 6$ for $n \equiv 0 \pmod{4}$ and $m \equiv 0 \pmod{4}$, $m \geq n \geq 5$.

Proof. We partition $G_{n,m}$ to a $G_{4i+1,4j+1}$ and the remain part which $i = \lfloor(n-1)/4\rfloor$ and $j = \lfloor(m-1)/4\rfloor$. The remaining part can be partition to a $G_{3,n-3}$, a $G_{3,m-3}$, and a $G_{3,3}$. Combining the results of Lemmas 3 and 4, we immediately obtain the upper bound $6\lfloor(n-1)/4\rfloor \lfloor(m-1)/4\rfloor + 2\lfloor(n-1)/4\rfloor + 2\lfloor(m-1)/4\rfloor + 3(m+n)/2 - 6$. $\square$

For example, Figure 7 shows a 2-rainbow domination of $G_{8,8}$ which can be partitioned to a $G_{5,5}$, two $G_{3,5}$ and a $G_{3,3}$.

Lemma 7 $\gamma_{r2}(G_{n,m}) \leq 6\lfloor(n-1)/4\rfloor \lfloor(m-1)/4\rfloor + 2\lfloor(n-1)/4\rfloor + 2\lfloor(m-1)/4\rfloor + \lfloor((m-1) \bmod 4)n/2\rfloor + \lfloor((n-1) \bmod 4)(m-(m-1) \bmod 4)/2\rfloor$ for $n \bmod 4 \neq 0$ and $m \bmod 4 \neq 0$ or $n \bmod 4 \neq 2$ and $m \bmod 4 \neq 2$, $m \geq n \geq 5$.

Proof. We partition $G_{n,m}$ to a $G_{4i+1,4j+1}$ and the remain part which $i = \lfloor(n-1)/4\rfloor$ and $j = \lfloor(m-1)/4\rfloor$. The remaining part can be partition to a $G_{(m-1) \bmod 4, n}$ and a $G_{(n-1) \bmod 4, m - (m-1) \bmod 4}$. Combining the results of Lemmas 3 and 4, we immediately obtain the upper bound $6\lfloor(n-1)/4\rfloor \lfloor(m-1)/4\rfloor + 2\lfloor(n-1)/4\rfloor + 2\lfloor(m-1)/4\rfloor + \lfloor((m-1) \bmod 4)n/2\rfloor + \lfloor((n-1) \bmod 4)(m-(m-1) \bmod 4)/2\rfloor$. $\square$

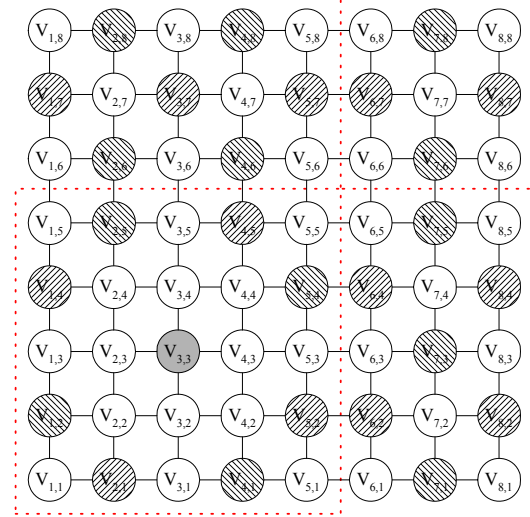


Figure 7: A 2-rainbow domination of $G_{8,8}$. The vertex with a color set $\{1, 2\}$ is represented by a gray vertex, and the vertex with color sets $\{1\}$ and $\{2\}$ are filled with slashes and backslashes, respectively.

As a concluding remark, we summarize Corollary 2, Lemmas 3, 5, 6, and 7 as follows.

Theorem 8 The upper bounds of $\gamma_{r2}(G_{n,m})$ are shown as below.

- (1) $\gamma_{r2}(G_{1,m}) = \lfloor m/2 \rfloor + 1$ for $m \geq 1$.
- (2) $\gamma_{r2}(G_{n,m}) \leq \lfloor n \times m/2 \rfloor$ for $2 \leq n \leq 4$ and $m \geq n$.
- (3) $\gamma_{r2}(G_{n,m}) \leq 6\lfloor(n-1)/4\rfloor \lfloor(m-1)/4\rfloor + 2\lfloor(n-1)/4\rfloor + 2\lfloor(m-1)/4\rfloor + (m+n)/2$ for $n \equiv 2 \pmod{4}$ and $m \equiv 2 \pmod{4}$, $m \geq n \geq 5$.
- (4) $\gamma_{r2}(G_{n,m}) \leq 6\lfloor(n-1)/4\rfloor \lfloor(m-1)/4\rfloor + 2\lfloor(n-1)/4\rfloor + 2\lfloor(m-1)/4\rfloor + 3(m+n)/2 - 6$ for $n \equiv 0 \pmod{4}$ and $m \equiv 0 \pmod{4}$, $m \geq n \geq 5$.
- (5) $\gamma_{r2}(G_{n,m}) \leq 6\lfloor(n-1)/4\rfloor \lfloor(m-1)/4\rfloor + 2\lfloor(n-1)/4\rfloor + 2\lfloor(m-1)/4\rfloor + \lfloor((m-1) \bmod 4)n/2\rfloor + \lfloor((n-1) \bmod 4)(m-(m-1) \bmod 4)/2\rfloor$ for $n \bmod 4 \neq 0$ and $m \bmod 4 \neq 0$ or $n \bmod 4 \neq 2$ and $m \bmod 4 \neq 2$, $m \geq n \geq 5$.

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